

Experiment 6: Grocery Purchase Cost Calculation

AIM

To calculate the total cost of a customer's purchase, including discounts and taxes, using arithmetic operations in Python.

ALGORITHM

1. Initialize the prices and quantities of grocery items.
2. Calculate the subtotal by multiplying each item's price by its quantity and adding the results.
3. Calculate the discount amount.
4. Subtract the discount from the subtotal.
5. Calculate the tax on the discounted amount.
6. Add the tax to the discounted amount.
7. Display the subtotal, discount, tax and final cost.

CODE

```
prices = [100, 200, 150]
quantities = [2, 1, 3]
discount_rate = 10
tax_rate = 5
subtotal = sum(
    prices[i] * quantities[i]
    for i in range(len(prices))
)
discount = subtotal * discount_rate / 100
discounted_total = subtotal - discount
tax = discounted_total * tax_rate / 100
total_cost = discounted_total + tax
print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total cost:", total_cost)
```

INPUT

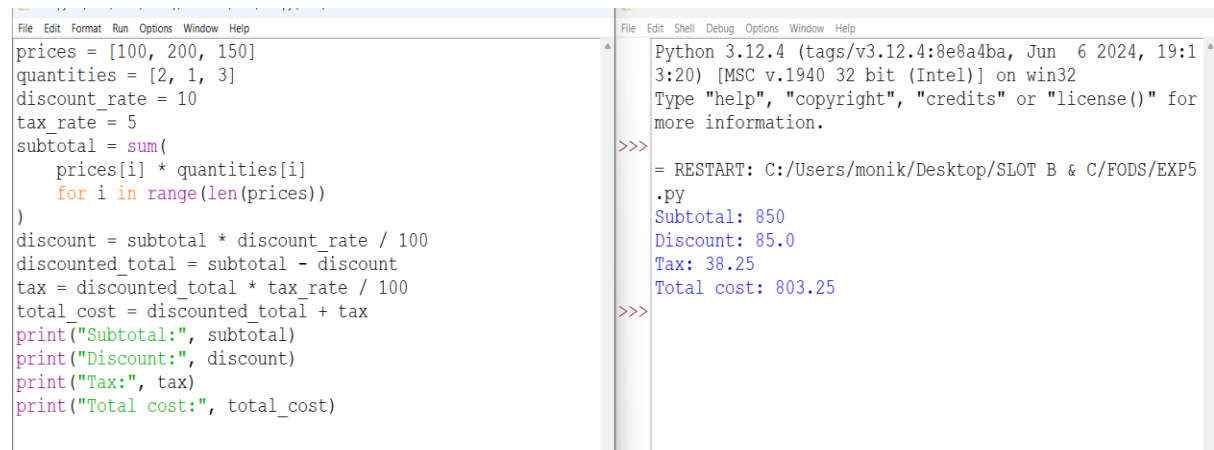
Prices = [100, 200, 150]

Quantities = [2, 1, 3]

Discount rate = 10%

Tax rate = 5%

OUTPUT

The image shows a screenshot of a Python IDE with two windows. The left window displays the Python code for calculating the total cost of a grocery purchase. The code defines prices, quantities, discount rate, and tax rate, then calculates the subtotal, discount, tax, and total cost. The right window shows the output of the code, which includes the Python version, the file path, and the calculated values for subtotal, discount, tax, and total cost.

```
File Edit Format Run Options Window Help
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total_cost = discounted_total + tax
print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total cost:", total_cost)
```

```
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:1
3:20) [MSC v.1940 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for
more information.
>>>
= RESTART: C:/Users/monik/Desktop/SLOT B & C/FODS/EXP5
.py
Subtotal: 850
Discount: 85.0
Tax: 38.25
Total cost: 803.25
>>>
```

RESULT

Thus, the total cost of the grocery purchase, including a 10% discount and 5% tax, was calculated as 803.25.